

Three Unifying Principles

Prediction Error

Content-Addressable Memory

Active Inference

WP1: Memory & CL

Distributed memory modules

- Surprise-gated write
- Hierarchical time scales
- Online single-pass learning

92.5% C_{ORE}50-NI
· 92.8% C_{LEAR}-10
94% assoc. recall · FGT $\approx$ 0

C_{ORE}50, Split-CIFAR, Path-X

Foundation

Design rules

PAM

Temporal Predictive Coding

HDC Kanerva Memory

Extension

Compositions

WP2: Reasoning

- Algebraic HDC composition
- FOMAML meta-learning
- Active inference
- Causal world models

40% ARC LOO
· 26% held-out
memory ablation: 30–43%

ARC, letter analogies, Ravens, bAbl

Stress test

WP3: Long-Horizon Agents

- Memory Maze ($T=10,000$) · Minecraft · Factorio
- 3-level policy hierarchy · Adaptive curriculum
- Causal interventions · Module specialisation analysis